KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Product, Quotient, and Chain Rules

Name _____

S1

Differentiate with respect to x.

$y = (3x + 4)^3(x - 2)^2$

S2

Differentiate with respect to x.

$y = (3x + 5)^4$

S3

Differentiate with respect to x.

$y = \dfrac{5x^2}{2x + 2}$

S4

Differentiate with respect to x.

$y = (x^2 + 3x)^3 \sin(2x)$

Show your working

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ALG-1258-L2 · S

Show your working

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Exit ticket

Product, Quotient and Chain Rule

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Name _____

E1

Differentiate with respect to x.

$y = (2x + 2)^4(x - 3)^2$

E2

Differentiate with respect to x.

$y = (5x + 5)^3$

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Differentiate with respect to x.

$y = \dfrac{3x^4}{4x + 3}$

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Differentiate with respect to x.

$y = (2x + 2)^4(x - 3)^2$

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Differentiate with respect to x.

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Differentiate with respect to x.

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Differentiate with respect to x.

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Show your working

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S1

Differentiate with respect to x.

$y = (3x + 4)^3(x - 2)^2$

Answer:

$\frac{dy}{dx} = 5(3x + 4)^2(x - 2)(3x - 2)$

Working:

$u = (3x + 4)^3, \quad v = (x - 2)^2$

$u' = 3(3x + 4)^{3-1} \cdot 3$

$u' = 9(3x + 4)^2$

$v' = 2(x - 2)^{2-1} \cdot 1$

$v' = 2(x - 2)$

$y' = u'v + uv'$

$y' = (9(3x + 4)^2)((x - 2)^2) + ((3x + 4)^3)(2(x - 2))$

Factor out the common factor $(3x + 4)^2(x - 2)$.

$y' = (3x + 4)^2(x - 2)[9(x - 2) + 2(3x + 4)]$

Expand and collect inside the bracket.

$y' = (3x + 4)^2(x - 2)[9x - 18 + 6x + 8]$

$y' = (3x + 4)^2(x - 2)(15x - 10)$

Take out the numerical factor 5 from the last bracket.

Write the factorised derivative.

$\frac{dy}{dx} = 5(3x + 4)^2(x - 2)(3x - 2)$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S2

Differentiate with respect to x.

$y = (3x + 5)^4$

Answer:

$\frac{dy}{dx} = 12(3x + 5)^3$

Working:

$u = 3x + 5, \quad y = u^4$

$\frac{dy}{du} = 4u^{4-1}$

$\frac{dy}{du} = 4u^3$

$\frac{du}{dx} = 3$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$\frac{dy}{dx} = 4u^3 \cdot 3$

$\frac{dy}{dx} = 4(3x + 5)^3 \cdot 3$

$\frac{dy}{dx} = 12(3x + 5)^3$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S3

Differentiate with respect to x.

$$y = \frac{5x^2}{2x+2}$$

Answer:

$$\frac{dy}{dx} = \frac{5x(x+2)}{2(x+1)^2}$$

Working:

$u = 5x^2, \quad v = 2x+2$

$u' = 5 \cdot 2x^{2-1}$

$u' = 10x^1$

$v' = 2$

$y' = \dfrac{u'v - uv'}{v^2}$

$u'v = (10x^1)(2x+2)$

$uv' = (5x^2)(2)$

$v^2 = (2x+2)^2$

$$\frac{dy}{dx} = \frac{10x^1(2x+2) - (5x^2)(2)}{(2x+2)^2}$$

Factor $5x$ out of the numerator.

$$y' = \frac{5x[2(2x+2) - 2x]}{(2x+2)^2}$$

Expand and collect inside the bracket.

$$y' = \frac{5x[4x+4-2x]}{(2x+2)^2}$$

$$y' = \frac{5x(2x+4)}{(2x+2)^2}$$

Take out the numerical factor 2 from the numerator bracket.

Use $2x+2 = 2(x+1)$ and reduce the constant fraction.

Write the factorised derivative.

$$\frac{dy}{dx} = \frac{5x(x+2)}{2(x+1)^2}$$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S4

Differentiate with respect to x.

$y = (x^2 + 3x)^3 \sin(2x)$

Answer:

$\frac{dy}{dx} = x^2(x+3)^2[3(2x+3)\sin(2x) + 2x(x+3)\cos(2x)]$

Working:

$u = (x^2 + 3x)^3, \quad v = \sin(2x)$

$w = x^2 + 3x, \quad w' = 2x + 3$

$u' = 3w^2 w'$

$u' = 3(x^2 + 3x)^2(2x + 3)$

$v' = \cos(2x) \cdot 2$

$v' = 2\cos(2x)$

$y' = u'v + uv'$

$y' = [3(x^2 + 3x)^2(2x + 3)]\sin(2x) + (x^2 + 3x)^3[2\cos(2x)]$

$\frac{dy}{dx} = 3(x^2 + 3x)^2(2x + 3)\sin(2x) + 2(x^2 + 3x)^3\cos(2x)$

Factor out $(x^2 + 3x)^2$, then use $x^2 + 3x = x(x + 3)$.

$\frac{dy}{dx} = x^2(x+3)^2[3(2x+3)\sin(2x) + 2x(x+3)\cos(2x)]$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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E1

Differentiate with respect to x.

$y = (2x + 2)^4(x - 3)^2$

Answer:

$\frac{dy}{dx} = 32(x + 1)^3(x - 3)(3x - 5)$

Working:

$u = (2x + 2)^4, \quad v = (x - 3)^2$

$u' = 4(2x + 2)^{4-1} \cdot 2$

$u' = 8(2x + 2)^3$

$v' = 2(x - 3)^{2-1} \cdot 1$

$v' = 2(x - 3)$

$y' = u'v + uv'$

$y' = (8(2x + 2)^3)((x - 3)^2) + ((2x + 2)^4)(2(x - 3))$

Factor out the common factor $(2x + 2)^3(x - 3)$.

$y' = (2x + 2)^3(x - 3)[8(x - 3) + 2(2x + 2)]$

Expand and collect inside the bracket.

$y' = (2x + 2)^3(x - 3)[8x - 24 + 4x + 4]$

$y' = (2x + 2)^3(x - 3)(12x - 20)$

Take out the numerical factor 4 from the last bracket.

Also use $2x + 2 = 2(x + 1)$ and collect the constant factors.

Write the factorised derivative.

$\frac{dy}{dx} = 32(x + 1)^3(x - 3)(3x - 5)$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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E2

Differentiate with respect to x.

$y = (5x + 5)^3$

Answer:

$\frac{dy}{dx} = 15(5x + 5)^2$

Working:

$u = 5x + 5, \quad y = u^3$

$\frac{dy}{du} = 3u^{3-1}$

$\frac{dy}{du} = 3u^2$

$\frac{du}{dx} = 5$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$\frac{dy}{dx} = 3u^2 \cdot 5$

$\frac{dy}{dx} = 3(5x + 5)^2 \cdot 5$

$\frac{dy}{dx} = 15(5x + 5)^2$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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E3

Differentiate with respect to x.

$$y = \frac{3x^4}{4x+3}$$

Answer:

$$\frac{dy}{dx} = \frac{36x^3(x+1)}{(4x+3)^2}$$

Working:

$u = 3x^4, \quad v = 4x+3$

$u' = 3 \cdot 4x^{4-1}$

$u' = 12x^3$

$v' = 4$

$y' = \dfrac{u'v - uv'}{v^2}$

$u'v = (12x^3)(4x+3)$

$uv' = (3x^4)(4)$

$v^2 = (4x+3)^2$

$$\frac{dy}{dx} = \frac{12x^3(4x+3) - (3x^4)(4)}{(4x+3)^2}$$

Factor $3x^3$ out of the numerator.

$$y' = \frac{3x^3[4(4x+3) - 4x]}{(4x+3)^2}$$

Expand and collect inside the bracket.

$$y' = \frac{3x^3[16x + 12 - 4x]}{(4x+3)^2}$$

$$y' = \frac{3x^3(12x+12)}{(4x+3)^2}$$

Take out the numerical factor 12 from the numerator bracket.

Write the factorised derivative.

$$\frac{dy}{dx} = \frac{36x^3(x+1)}{(4x+3)^2}$$